Report Lab - 2

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Commands used

nmap -sn 192.168.100.248/24 : For discovering active hosts (figure 1)

nmap 192.168.100.248 : For scan open ports (figure 2)

nmap -sV 192.168.100.248 : For services detection (figure 3)

sudo nmap -O 192.168.100.248 : For Os detection (figure 4)

sudo nmap -A 192.168.100.248 : Used for agressive scan (figure 5)

Screenshots

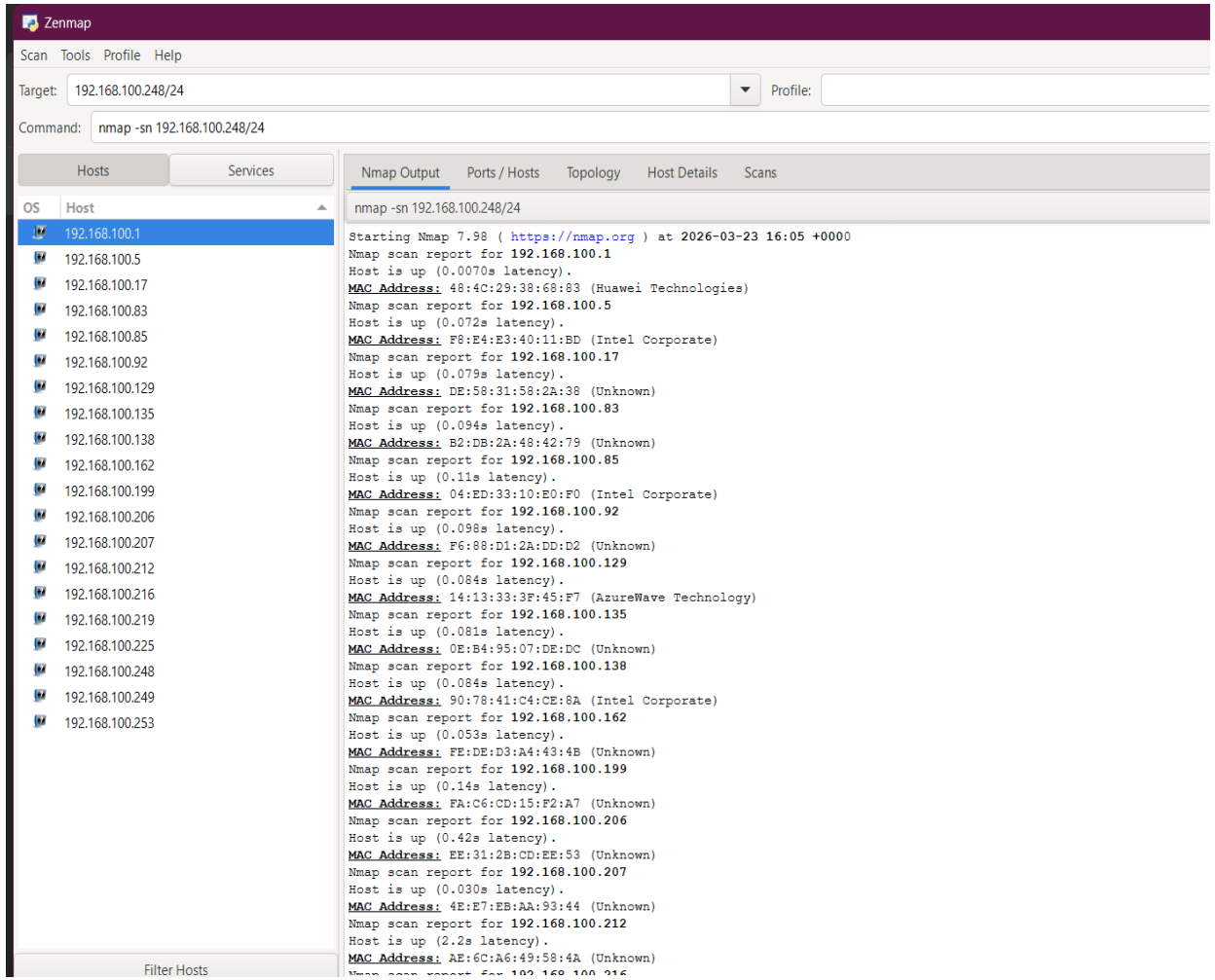

Figure 1

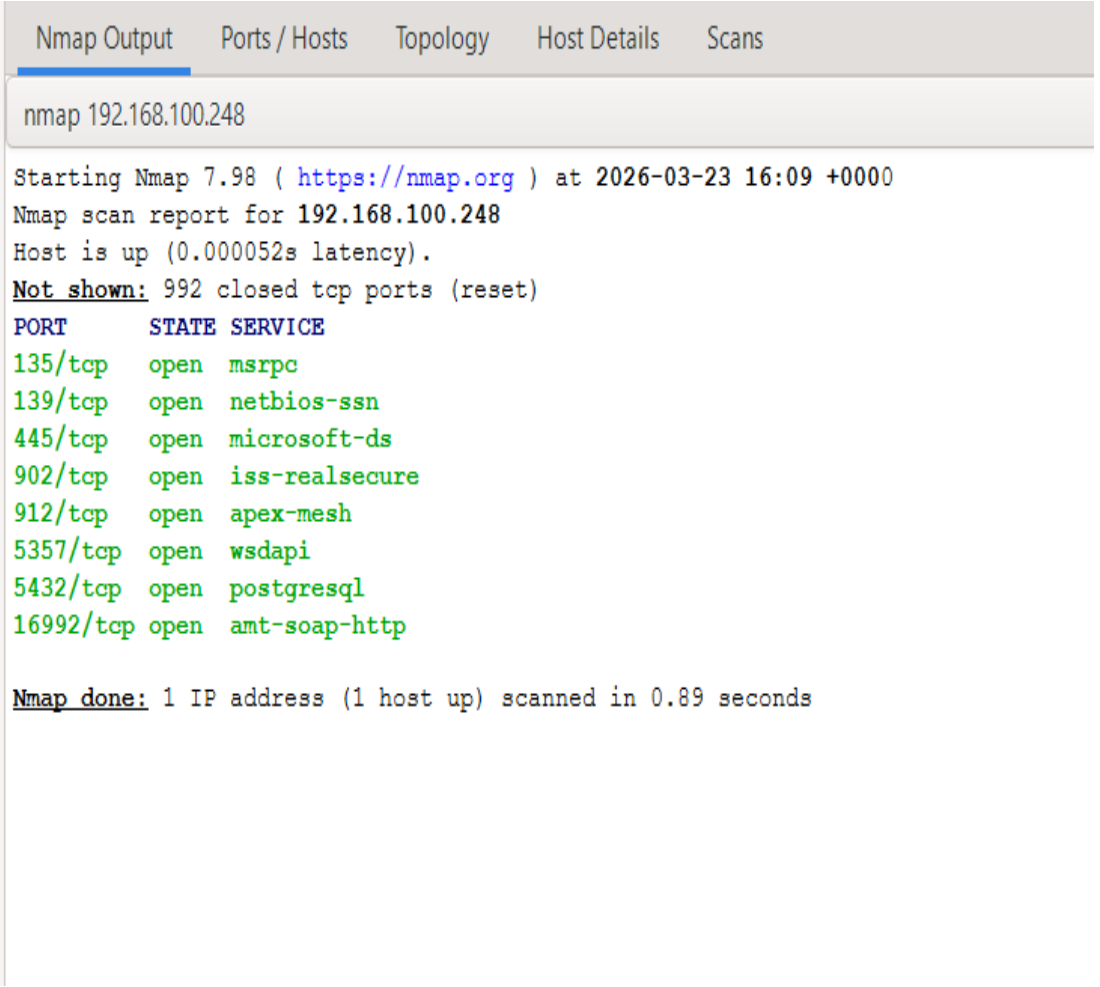

Figure 2

Screenshots (next)

Nmap Output | Ports / Hosts | Topology | Host Details | Scans

nmap -sV 192.168.100.248

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:11 +0000
Nmap scan report for 192.168.100.248
Host is up (0.00088s latency).
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE         VERSION
135/tcp   open  msrpc           Microsoft Windows RPC
139/tcp   open  netbios-ssn     Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
902/tcp   open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
912/tcp   open  vmware-auth     VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
5357/tcp  open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
5432/tcp  open  postgresql      PostgreSQL DB
16992/tcp open  http            Intel Active Management Technology User Notification Service httpd 12.1.1.1022
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:12.1.1.1022

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.33 seconds
```

Figure 3

Nmap Output | Ports / Hosts | Topology | Host Details | Scans

nmap -O 192.168.100.248

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:13 +0000
Nmap scan report for 192.168.100.248
Host is up (0.00078s latency).
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
902/tcp   open  iss-realsecure
912/tcp   open  apex-mesh
5357/tcp  open  wsdapi
5432/tcp  open  postgresql
16992/tcp open  amt-soap-http
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.98%E=4%D=3/23%OT=135%CT=1%CU=34308%PV=Y%DS=0%DC=L%G=Y%TM=69C166
OS:CB%P=i686-pc-windows-windows)SEQ(SP=101%GCD=1%ISR=10E%TI=I%CI=I%II=I%SS=
OS:S%TS=A)SEQ(SP=104%GCD=1%ISR=10A%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=104%GCD=
OS:2%ISR=10D%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=107%GCD=1%ISR=10B%TI=I%CI=I%II
OS:=I%SS=S%TS=A)SEQ(SP=FD%GCD=1%ISR=105%TI=I%CI=I%II=I%SS=S%TS=A)OPS(O1=MFF
OS:D7NW8ST11%O2=MFFD7NW8ST11%O3=MFFD7NW8NNT11%O4=MFFD7NW8ST11%O5=MFFD7NW8ST
OS:11%O6=MFFD7ST11)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFFF)ECN(
OS:R=Y%DF=Y%T=80%W=FFFF%O=MFFD7NW8NNS%CC=N%Q=)T1(R=Y%DF=Y%T=80%S=O%A=S+%F=A
OS:S%RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S%F=AR%O=%RD=0%Q=)T3(R=Y%DF=Y%T=80%
OS:W=0%S=Z%A=O%F=AR%O=%RD=0%Q=)T4(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=0%Q=)
OS:T5(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=80%W=0%S=A%A
OS:=O%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%D
OS:F=N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=8
OS:0%CD=Z)

Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.10 seconds
```

Figure 4

Screenshots (next)

Nmap Output | Ports / Hosts | Topology | Host Details | Scans

nmap -A 192.168.100.248

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:15 +0000
Nmap scan report for 192.168.100.248
Host is up (0.00054s latency).
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE         VERSION
135/tcp   open  msrpc           Microsoft Windows RPC
139/tcp   open  netbios-ssn     Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
902/tcp   open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
912/tcp   open  vmware-auth     VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
5357/tcp  open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Service Unavailable
5432/tcp  open  postgresql      PostgreSQL DB
16992/tcp open  http            Intel Active Management Technology User Notification Service httpd 12.1.1.1022
| http-title: Intel&reg; Active Management Technology
|_Requested resource was /logon.htm
|_http-server-header: Intel(R) Active Management Technology 12.1.1.1022
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.98%E=4%D=3/23%OT=135%CT=1%CU=35781%PV=Y%DS=0%DC=L%G=Y%TM=69C167
OS:50%P=i686-pc-windows-windows)SEQ(SP=100%GCD=1%ISR=10C%TI=I%CI=I%II=I%SS=
OS:S%TS=A)SEQ(SP=103%GCD=1%ISR=10C%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=105%GCD=
OS:1%ISR=108%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=F4%GCD=1%ISR=108%TI=I%CI=I%II=
OS:I%SS=S%TS=A)SEQ(SP=FB%GCD=1%ISR=10F%TI=I%CI=I%II=I%SS=S%TS=A)OPS(O1=MFFD
OS:7NW8ST11%O2=MFFD7NW8ST11%O3=MFFD7NW8NNT11%O4=MFFD7NW8ST11%O5=MFFD7NW8ST1
OS:1%O6=MFFD7ST11)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFFF)ECN(R
OS:=Y%DF=Y%T=80%W=FFFF%O=MFFD7NW8NNS%CC=N%Q=)T1(R=Y%DF=Y%T=80%S=O%A=S+%F=AS
OS:%RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S%F=AR%O=%RD=0%Q=)T3(R=Y%DF=Y%T=80%W
OS:=0%S=Z%A=O%F=AR%O=%RD=0%Q=)T4(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=0%Q=)T
OS:5(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=80%W=0%S=A%A=
OS:O%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF
OS:=N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=80
OS:%CD=Z)

Network Distance: 0 hops
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:12.1.1.1022

Host script results:
| smb2-time:
|   date: 2026-03-23T16:16:02
|_  start_date: N/A
| smb2-security-mode:
|   3.1.1:
```

Figure 5

Identified hosts and services

Identified host:

192.168.100.1

192.168.100.5

192.168.100.83

192.168.100.92

Services

http

microsoft-ds

msrpc

netbios-ssn

postgresql

vmware-auth

Answering to the questions

How many hosts were detected ?

15 hosts were detected during the detection

Which ports are most common ?

The most common ports are : 135, 139, 445 for Microsoft services

Why are open ports a risk ?

Every open port is a potential entry point. If the service listening on that port has a vulnerability, an attacker can use it to take control of the system.

What information can attackers gather ?

They can identify the operating system, the precise versions of the software, and even the system time.

Security observation

Attack surface: Too many administration services (VMware, Intel AMT) are exposed.

SMB vulnerability: The lack of mandatory signing on port 445 facilitates man-in-the-middle attacks.

Data exposure: The PostgreSQL database (5432) is accessible without apparent network restriction.